Solution to the Daily Limit

Based on the problem provided in the image dailyintegral-limit-limit-1x1-2026-08-11.jpg, we are tasked with finding the limit of the sequence $(u_n)_{n\in\mathbb{N}^*}$, defined as:

$$\forall n \in \mathbb{N}^*, \quad u_n = \left(\prod_{k=1}^{n}\left(1+\frac{k}{n}+\frac{k}{n^2}\right)\right)^{\frac{1}{n}}$$

$$\text{Find: } \lim_{n\to\infty} u_n$$

1. Taking the Natural Logarithm

To evaluate the limit of a sequence involving an n-th root of a product, it is standard to first examine the natural logarithm of the sequence, $\ln(u_n)$:

$$\ln(u_n) = \frac{1}{n}\sum_{k=1}^{n}\ln\left(1+\frac{k}{n}+\frac{k}{n^2}\right)$$

We can factor the argument of the logarithm to isolate the dominant terms that will form our Riemann sum:

$$1+\frac{k}{n}+\frac{k}{n^2} = \left(1+\frac{k}{n}\right)\left(1+\frac{\frac{k}{n^2}}{1+\frac{k}{n}}\right)$$

Using the property of logarithms, $\ln(ab) = \ln(a)+\ln(b)$, we split the summation into two distinct parts:

$$\ln(u_n) = \frac{1}{n}\sum_{k=1}^{n}\ln\left(1+\frac{k}{n}\right) + \frac{1}{n}\sum_{k=1}^{n}\ln\left(1+\frac{k}{n^2+nk}\right)$$

2. Evaluating the Error Term

Let us analyze the second sum as $n\to\infty$. For $1\le k\le n$, we can bound the fraction:

$$\frac{k}{n^2+nk} \le \frac{n}{n^2+0} = \frac{1}{n}$$

Since $\ln(1+x)\le x$ for all $x\ge 0$, we can establish an upper bound for the terms in the second sum:

$$0 \le \frac{1}{n}\sum_{k=1}^{n}\ln\left(1+\frac{k}{n^2+nk}\right) \le \frac{1}{n}\sum_{k=1}^{n}\frac{1}{n} = \frac{1}{n^2}\cdot n = \frac{1}{n}$$

By the Squeeze Theorem, as $n\to\infty$, the upper bound $\frac{1}{n}\to 0$. Therefore, the second sum converges to 0.

3. Evaluating the Riemann Sum

With the error term vanishing, the limit of $\ln(u_n)$ is entirely determined by the first sum, which is a standard Riemann sum:

$$\lim_{n\to\infty} \ln(u_n) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \ln\left(1 + \frac{k}{n}\right)$$

This limit represents the definite integral of the function $f(x) = \ln(1+x)$ over the interval $[0,1]$. We can evaluate this integral using integration by parts (letting $w = 1+x$, $dw = dx$):

$$\begin{aligned}\int_0^1 \ln(1+x)\,dx &= \Big[(1+x)\ln(1+x) - x\Big]_0^1 \\ &= \Big(2\ln(2) - 1\Big) - \Big(1\ln(1) - 0\Big) \\ &= 2\ln(2) - 1\end{aligned}$$

Conclusion

We have established that $\lim_{n\to\infty} \ln(u_n) = 2\ln(2) - 1$. To find the limit of the original sequence u_n, we exponentiate the result:

$$\begin{aligned}\lim_{n\to\infty} u_n &= e^{2\ln(2)-1} \\ &= e^{\ln(4)-1} \\ &= \frac{e^{\ln(4)}}{e} \\ &= \frac{4}{e}\end{aligned}$$

Thus, the final limit of the sequence is:

$$\lim_{n\to\infty} u_n = \frac{4}{e}$$